Young Innovators · Class 20

ESP32-CAM Wildlife Camera

Who visits our school garden when nobody is looking?

Pixels | Wi-Fi stream | Camera trap



Did you know?
Camera traps photographed snow leopards in Gilgit-Baltistan, helping scientists count this rare cat.

LEARN

Let's think...

How do you photograph a shy animal?

Nobody home

Birds, cats and squirrels vanish the moment a person walks out. Scientists can't sit in the forest for weeks. So how do they get thousands of photos of animals that hide from humans?



Hide in a bush?

Zoom lens?

A camera that waits!

Today: you build a **camera trap**: an ESP32-CAM that streams over Wi-Fi and snaps a photo to SD whenever something moves.

Learning objectives

By the end of this class I can...

1	**How cameras see:** explain **pixels**, **resolution** and why photos are saved as **JPEG**	**Proof:** I can estimate how many photos fit on an SD card
2	**Program the ESP32-CAM:** upload code to an ESP32-CAM and **stream** live video to a phone over Wi-Fi	**Proof:** I watched my camera's stream in a browser
3	**Camera trap:** use a **PIR** to take a photo when something moves and save it to **microSD**	**Proof:** my trap saved a photo of a 'visitor'

Objective 3 · Camera trap · video

Video time!



The Art & Science Of Camera Trapping

Wildlife Conservation Trust · about 4 minutes · youtu.be/Mnw8F0HAyBA

Watch for...

1. How does a camera trap know when to take a photo?
2. Where do scientists **place** the traps, and why?
3. What can scientists learn from the photos?

Scan to watch

After the video: Which animal would you most like to photograph at our school, and where would you place the trap?

Objective 1 · How cameras see

Pixels, resolution and JPEG

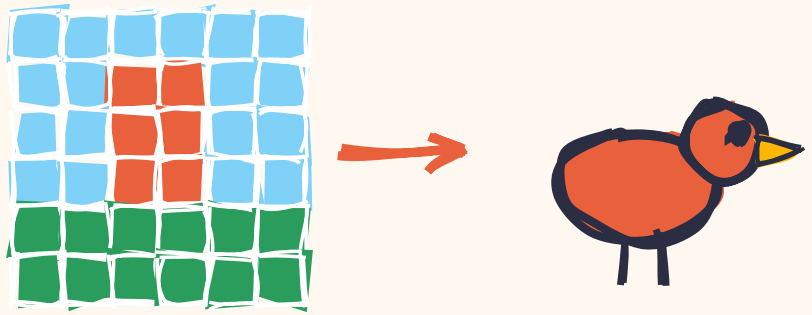
The idea

The camera chip (OV2640) is a grid of tiny light sensors: one per **pixel**. **UXGA** = 1600 × 1200 = about **2 million pixels**.

Raw, each pixel needs 3 bytes (R, G, B): 1600 × 1200 × 3 ≈ **5.8 MB**! **JPEG** compresses it to about **150 to 300 kB** by throwing away detail our eyes don't notice.

16 GB card ÷ 200 kB ≈ **80 000 photos**.

Example: a mosaic of tiles



A wall mosaic made of 2 million tiles: each tile is one colour. Stand back and you see a bird. JPEG is like describing big same-colour areas in one sentence instead of tile by tile.

Objective 3 · Camera trap

A camera trap waits, then wakes

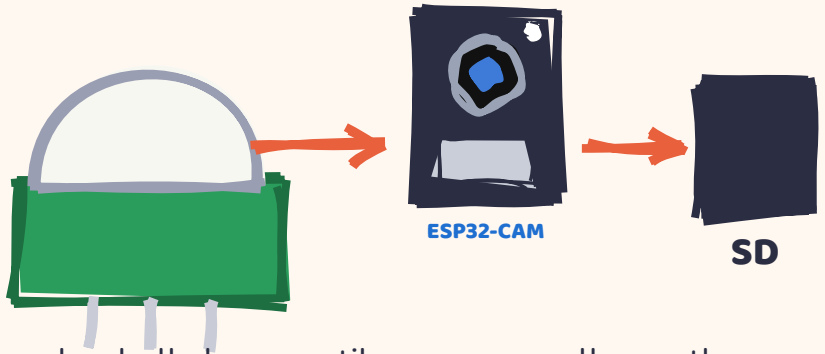
The idea

A **PIR** sensor sees **body heat moving**. When it fires: take a photo, save it, wait a few seconds, repeat.

The trap must run for days on a battery, so real traps **sleep** most of the time and only wake on motion.

Wildlife rule: no bait that harms animals, no flash in their eyes at night, never disturb nests.

Example: a doorbell camera



A video doorbell sleeps until someone walks up, then records. Our trap is a doorbell for birds and cats!

Objective 2 · Program the ESP32-CAM

Uploading to an ESP32-CAM

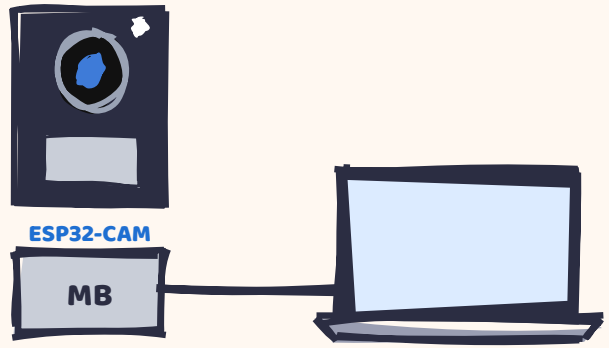
The idea

The AI-Thinker ESP32-CAM has **no USB port**. We use a programmer board (ESP32-CAM-MB, or an FTDI adapter).

To upload: hold **IO0** to GND (MB board: hold the IO0 button), press **RST**, click Upload.
To run: release IO0, press RST again.

Board: **AI Thinker ESP32-CAM**. Power it with a good **5 V**: weak power = brownout resets.

Example: a shop's back door



Deliveries (new code) go through the back door, which is only open when the manager (IO0) says so. Normal customers use the front door: running mode.

Real world

Cameras for nature



Snow leopard surveys
Camera traps in Pakistan's northern mountains



Counting species
Scientists ID animals from millions of photos with AI



Anti-poaching
Traps send alerts when people enter protected areas



Bird feeders
Smart feeders name the bird that visited



What animals do you think visit our school at night?

AI helpers
Projects like Snapshot Serengeti use volunteers and AI to label millions of trap photos.

Quick quiz

Camera maths

Question	Answer
1. 640 × 480 = how many pixels?	
2. Why save as JPEG, not raw?	
3. Each photo is 100 kB. How many fit in 1 GB?	
4. What does the PIR detect?	
5. ESP32-CAM keeps resetting. First thing to check?	

Answers

Cover! 1: 307 200 · 2: about 20× smaller · 3: about 10 000 · 4: moving body heat · 5: the 5 V power supply

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Respect animals and people
2. Human camera trap
3. Our camera kit (per group)
4. Stream live video
5. Motion-triggered photos to SD
6. Set the trap
7. Field log

MAKE

Before we start

Respect animals and people

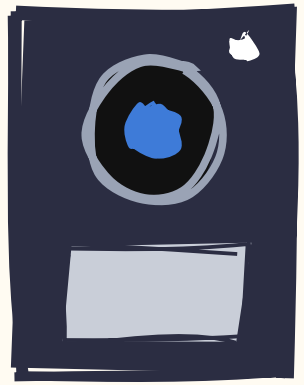
Animals first Never disturb nests or trap animals. Observe only.

Privacy Point the camera at the garden, not at classrooms, toilets or neighbours' houses.

Permission Ask the school before placing a camera, and put up a small 'camera trap' sign.

Weather Electronics + rain = damage. Use a box with a clear window and a roof.

Network Don't share your camera's IP address or stream link publicly.



ESP32-CAM

Ethics

Good scientists collect only the data they need, and protect what they collect.

MAKE

Class game

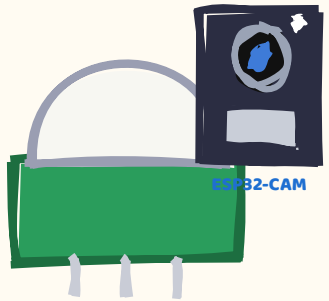
Human camera trap

How to play

1. One student is the **camera**, eyes closed, holding a phone (no photos!) or a paper frame.
2. Another is the **PIR**, watching the door. When anyone crosses, the PIR taps the camera.
3. Camera opens eyes for 1 second and 'snaps': says what it saw.
4. Others try to sneak past slowly. Can they fool the PIR?

Then discuss

Who got past? Was it speed, or something else? (Hint: PIR likes movement across its view.)



Big idea

Real traps miss things too. Placement and angle matter.

Build it

Our camera kit (per group)



ESP32-CAM (AI-Thinker)
Camera + Wi-Fi + SD slot



ESP32-CAM-MB
USB programmer board



PIR (HC-SR501)
Motion trigger, output → IO13



microSD (≤ 16 GB, FAT32)
Stores the photos



5 V power bank
For the garden test



Box + clear window
Weather shield

Arduino IDE 2 with the esp32 boards package, board 'AI Thinker ESP32-CAM'. The class folder has cam_setup.h.

Hot chip
The ESP32-CAM gets warm when streaming. That's normal; hot to touch means unplug and check wiring.

MAKE

Objective 2 · Build step 1 of 3

Stream live video

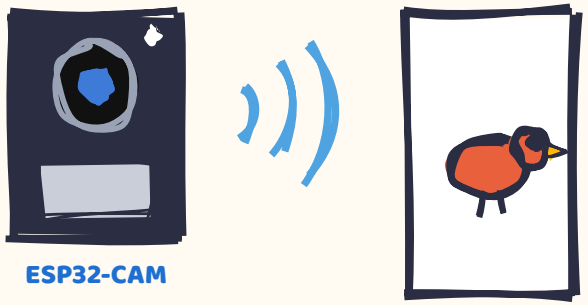
a File → Examples → ESP32 → Camera → **CameraWebServer**.

b Select **#define CAMERA_MODEL_AI_THINKER** (comment out the others).

c Type your Wi-Fi **ssid** and **password** (use the class hotspot).

d Hold IO0, press RST, Upload. Then release IO0, press RST, open Serial Monitor at 115200.

e Open the printed IP address on a phone on the same Wi-Fi. Press **Start Stream**!



Check: You can see the live camera on your phone.

Brownout detector message? Use a better USB cable or power supply.

MAKE

Objective 3 · Camera trap · Build step 2 of 3

Motion-triggered photos to SD

```ino
#include "esp_camera.h"
#include "SD_MMC.h"
#include "cam_setup.h"   // class file: AI-Thinker pins + startCamera()

const int PIR = 13;
int photoNum = 0;

void setup() {
  Serial.begin(115200);
  startCamera();                    // from cam_setup.h
  SD_MMC.begin("/sdcard", true);    // 1-bit mode frees IO13
  pinMode(PIR, INPUT);
}

void loop() {
  if (digitalRead(PIR) == HIGH) {
    camera_fb_t *fb = esp_camera_fb_get();      // take a photo
    if (fb) {
      String path = "/pic" + String(photoNum++) + ".jpg";
      File f = SD_MMC.open(path, FILE_WRITE);
      f.write(fb->buf, fb->len);                // save JPEG bytes
      f.close();
      esp_camera_fb_return(fb);                 // free memory
      Serial.println("Saved " + path);
    }
    delay(5000);                                // 5 s between photos
  }
}
```

fb

fb is the frame buffer: the photo in memory. fb->len is its size in bytes.

Return it

Always esp_camera_fb_return(fb), or memory runs out after a few photos.

Wiring

PIR VCC → 5V, GND → GND, OUT → IO13. Card in before power-on.

MAKE

Objective 3 · Build step 3 of 3

Set the trap

a Test indoors: wave at the PIR. Check Serial: 'Saved /pic0.jpg'.

b Put the card in a laptop. Open the photos. Sharp? Pointing the right way?

c Place the box in the garden, 1 m off the ground, facing a feeder or path, **not** the sun.

d Power it from the power bank. Leave it for a break, or overnight with permission.

e Collect the card. Count and label the visitors: bird, cat, human, leaf blowing…



Check: You have photos and a tally of who visited.

Never climb or place the camera anywhere unsafe. Ask an adult.

Test like an engineer

Field log

Measure	Result	Notes
Photos taken		
Animal photos		
False triggers (wind, sun)		
Best photo: what is it?		
Battery time before it died		h

Debug checklist

SD card failed? Format FAT32, ≤ 16 GB, insert before power.
Black photos? Lens cap? Too dark?
Green tint on first photo? Take and throw away 1 photo after startCamera().

Scientists count false triggers too. A trap that saves 300 photos of leaves is a design problem to solve!

Level up!

Field scientist challenges

Choose your challenge



Deep sleep: sleep until the PIR wakes the chip (ext0 wakeup on IO13) to last for days.



Alert: send a Telegram or email message with the photo.



AI label: upload photos to Teachable Machine and train 'bird / cat / empty'.



Night: add an infrared LED (invisible to animals) for night photos.

Think like an engineer

Why does a flash at night disturb animals, but infrared does not?

Boss level

Run a 1-week survey of the school garden and present a chart of visitors by time of day.

Objective check

Can I really do it? Prove it.

01 A photo is 1600 × 1200 pixels. How big is it raw? Why is the JPEG much smaller?

02 Show your live stream on a phone. How did you put the board into upload mode?

03 Show a photo your trap saved. What triggered it?

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Photo gallery: each group shows its best visitor and one false trigger, and explains why it happened.

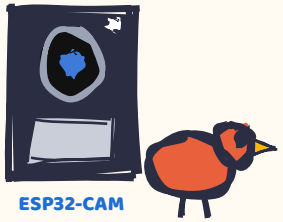
Show what you know

What did we learn?

- [x] Photos are grids of pixels; JPEG shrinks them
- [x] ESP32-CAM: hold IO0 to upload, stream over Wi-Fi
- [x] PIR detects moving body heat
- [x] fb_get → save to SD → fb_return
- [x] Respect animals, privacy and permission

At home

Watch your window for 10 minutes. Tally every animal visitor you see.



Exit ticket: why do we wait 5 seconds after each photo?

Next class:
Python Data Lab: Is Our Classroom Too Hot?